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CRITERION VII

INSTITUTIONAL VALUES AND BEST PRACTICES

7.1.9 CONSTITUTIONAL OBLIGATIONS

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Blockchain based E-Voting System on Smartphones

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Abstract - India, the world's largest democracy, faces significant challenges in conducting free and fair elections due to the prevalence of vote tampering, booth capturing, and rigging. To address and solve these challenges, a blockchain-based e-voting system will be developed that is secure, transparent, and accessible to all eligible voters in India. The decentralized nature of blockchain, combined with the transparency and security of smart contracts written in Solidity, will ensure that the voting process is tamper-proof and transparent. The system will be developed and tested on Ethereum testnet, and the complexity of existing systems will be simplified into a simple smartphone application that can be accessed by all eligible voters in India. The system will remove dependency on crypto wallets like Metamask, the backend api's are developed using node.js, express.js and web3.js. SHA-3 will be used for hashing and encryption purposes. The frontend will be developed using React Native, ensuring a user-friendly and accessible interface. The successful implementation of this blockchain-based e-voting system could revolutionize the democratic process in India, making it more accessible, secure, and transparent, and setting an example for other countries to follow.

Keywords – Blockchain, Ethereum, Decentralized, Solidity, Smart Contracts, Testnets, Metamask, React-Native, web3, Node.js, SHA-3, E-voting.

I. INTRODUCTION

The emergence of blockchain technology has opened up new possibilities for secure and transparent electronic transactions, including e-voting systems. However, many people are still unfamiliar with the technology and may find it challenging to understand the complexities of crypto wallets like Metamask, which are often required to participate in blockchain-based systems.

To overcome this challenge, the blockchain-based e-voting system on smartphones that we propose eliminates the dependency on people to download and set up a crypto wallet. Instead, the system randomly allocates the client with a key from the array of private keys, in the backend. It leverages the existing infrastructure of smartphones and mobile applications to provide a user-friendly interface that is easy to understand and use. This approach allows people

to participate in the voting process without the need for technical expertise or complex software installations.

The system uses a decentralized ledger to ensure transparency and integrity of the voting process, providing a tamper-proof and secure environment that is resistant to fraud and manipulation. The decentralized nature of the system means that there is no single point of failure, reducing the risk of cyberattacks and other security threats.

The use of blockchain technology also ensures anonymity and privacy for the voters, as their identities are protected by cryptographic algorithms. The system uses encryption and hashing algorithms to protect the data and prevent unauthorized access, ensuring the confidentiality and integrity of the voting process.

The choice of smartphones as the platform for the e-voting system is also significant, as it provides a ubiquitous and accessible platform that is widely used by people worldwide. The user-friendly interface of smartphones, combined with the secure and transparent nature of blockchain technology, makes it possible to develop a robust and reliable e-voting system that can be used in various political contexts.

The blockchain-based e-voting system on smartphones that we propose is designed to overcome the challenges and limitations of traditional voting systems while ensuring transparency, security, and efficiency.

II. LITERATURE REVIEW

[1] Michael A. Specter, James Koppel, and Daniel Weitzner, MIT presented a paper on *The Ballot is Busted Before the Blockchain: A Security Analysis of Voatz, the First Internet Voting Application Used in U.S. Federal Elections*

The paper analyzes the security of Voatz, a mobile voting application used in the 2018 U.S. federal elections, and identifies multiple vulnerabilities that could allow attackers to compromise the integrity and confidentiality of the voting process. The researchers found that the application was susceptible to attacks such as vote manipulation, vote buying, and denial-of-service attacks. The paper also questions the use of blockchain technology in online voting

systems and calls for greater transparency and public oversight of such systems. The study highlights the need for rigorous security testing and evaluation of online voting systems before their deployment in critical elections.

[2] Prof. Mrunal Pthak, Ajinkya Kadam, Amol Suradkar, Prashant Parde, Akansha Ghodeswar presented a paper on Blockchain-Based E-Voting System

The paper proposes a blockchain-based e-voting system that utilizes smartphones as a platform to overcome the challenges of traditional voting systems. The system uses a decentralized ledger to ensure transparency and integrity of the voting process, providing a tamper-proof and secure environment that is resistant to fraud and manipulation. The system also ensures anonymity and privacy for the voters by protecting their identities with cryptographic algorithms. The choice of smartphones as the platform for the e-voting system provides a user-friendly interface that is easy to use and accessible to people worldwide. Overall, the proposed system offers a robust and reliable alternative to traditional voting systems while ensuring transparency, security, and efficiency.

[3] Ehab Zaghloul; Tongtong Li; Jian Ren presented a paper on d-BAME: Distributed Blockchain-Based Anonymous Mobile Electronic Voting

The paper proposes d-BAME, a new design for a distributed blockchain-based anonymous mobile electronic voting system. The system leverages the blockchain technology to ensure transparency and security in the voting process while maintaining the anonymity of the voters. The design includes multiple layers of security mechanisms, including encryption and secure communication protocols, to ensure the integrity and confidentiality of the voting data. The proposed system is scalable, efficient, and easily deployable, making it suitable for various political contexts. The authors conducted a simulation-based evaluation of the proposed system, which demonstrated its effectiveness in ensuring secure and anonymous voting.

[4] Akhil Shah, Nishita Sodhia, Shruti Saha, Soumi Banerjee, Madhuri Chavan presented a paper on Blockchain Enabled Online-Voting System

The paper proposes a blockchain-enabled online voting system that provides a secure and transparent environment for conducting elections. The system uses a combination of blockchain technology, smart contracts, and cryptography to ensure the integrity and privacy of the voting process. The system allows voters to cast their votes securely from any location using their mobile devices, while the use of blockchain technology ensures transparency and immutability of the voting data. The system also provides mechanisms for verifying voter identity and preventing double voting, while the use of cryptography ensures the confidentiality and integrity of the voting process.

III. PRELIMINARIES

[1] Blockchain - Ethereum

Blockchain is a distributed ledger technology that allows multiple parties to share a digital database that is tamper-proof and transparent. It works by using cryptography to validate and secure transactions, which are then added to a block that is linked to previous blocks in a chain. Ethereum is a blockchain platform that enables developers to create and execute decentralized applications. It provides a secure and transparent environment for transactions and enables the use of smart contracts, which are self-executing contracts that can automate the execution of an agreement.

In the context of our research paper, we are proposing the use of blockchain technology (specifically Ethereum) as a means of securing and verifying the votes cast in an e-voting system on smartphones. By using blockchain, you can ensure that the votes are tamper-proof and transparent, as each vote will be recorded on the blockchain and linked to previous blocks in a chain. This can help to increase the trust and transparency of the e-voting system and reduce the risk of fraud or manipulation.

[2] Smart Contracts - Solidity

Smart contracts are self-executing contracts that automate the execution of an agreement between two parties. They are typically executed on a blockchain platform and use code to define the terms of the agreement. Solidity is a programming language used to write smart contracts on the Ethereum blockchain. It is specifically designed for writing smart contracts and allows developers to create complex contracts that can handle multiple conditions and execute automatically.

In the context of your research paper, you are proposing the use of smart contracts (written in Solidity) to automate the voting process in your e-voting system. The smart contract will define the rules and conditions of the voting process and automatically execute the results of the election. By using smart contracts, you can eliminate the need for intermediaries and increase the efficiency and accuracy of the voting process.

[3] React Native - Cross Platform Application

React Native is a framework for building cross-platform mobile applications using the React framework for building user interfaces. It allows developers to write code once and deploy it on both iOS and Android devices. This can help to reduce development costs and increase the speed of development, as developers only need to write code once for both platforms.

In the context of your research paper, you are proposing the use of a cross-platform mobile application (built using React Native) as the interface for your e-voting system. This will allow voters to cast their votes from their smartphones, which can increase accessibility and convenience for voters. By using React Native, you can ensure that the application runs smoothly on both iOS and Android devices and provide a seamless user experience for voters.

IV. PROPOSED SYSTEM

All software based applications follow a tech stack of frameworks, libraries, servers and database systems to formulate the core working of the application. We have incorporated ethereum as our database, particularly Sepolia which is an ethereum based test-network following all the protocols of the ethereum mainnet except the fact that the currency used in Sepolia is fake ETH. It is used to demonstrate how the transaction would actually take place in the mainnet. Using solidity we write smart contracts - Voters.sol and VotingPortal.sol to define the state of our data in the Sepolia database. The Voters.sol consists of data of verified voters, the functions in this contract are used to retrieve this data from the database and update their voting status once they have voted. The VotingPortal.sol is used to store the state of the candidates and define the rules and conditions of the voting process and its execution. Once these contracts are deployed, the server for our application is set up using express.js and web3.js, these frameworks and libraries are used to create API endpoints to interact and call functions and variables of the contract so as to retrieve as well as update the state of data in blockchain. The client and admin interfaces are provided with login systems which on verification allows the users to interact with the UI which is integrated with the server side backend. Hence providing a seamless user experience for voters and administrators.

[1] Database - Ethereum Testnet (Sepolia)

The proposed e-voting system utilizes the Ethereum testnet, Sepolia, as the database for storing and managing voting data. Sepolia is chosen because it follows all the protocols of the Ethereum mainnet, ensuring immutability and transparency in the voting process. The database uses smart contracts written in Solidity, such as the Voters.sol and VotingPortal.sol, to define the state of data in the Sepolia database. The Voters.sol contract contains data on verified voters, and functions in this contract are used to retrieve this data from the database and update their voting status once they have voted. The VotingPortal.sol contract stores the state of the candidates, defines the rules and conditions of the voting process, and executes the results of the election. This contract also keeps track of voterIDs that have voted while maintaining their anonymity.

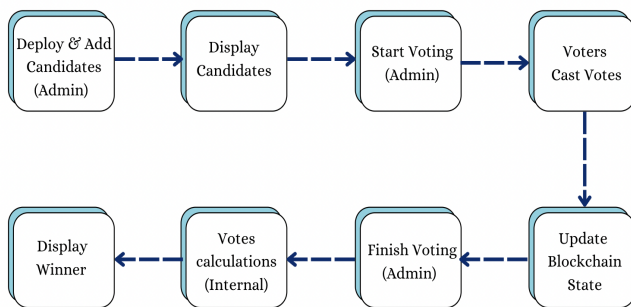


Fig 1: VotingPortal Contract Flow Diagram



Fig 2: Voters Contract Flow Diagram

[2] Server - Express & Web3

The server for the proposed system is set up using the express.js framework and web3.js library. The OTP (One-Time Password) system backend using Twilio is used for login verification, which adds an additional layer of security to the system. The voterIDs are encrypted using sha3 to maintain anonymity. An array of private keys is provided in the .env file, which means that users do not need to connect their wallet to the application. The web3js library is used for communication with the smart contract using the contract address and ABI (Application Binary Interface).

[3] Client Interface - React Native

The client interface is implemented using Expo, a platform for building cross-platform applications. The axios library is used for communicating with the server APIs. Redux is used for local state management of the user, which helps in managing the state of the application. A local authentication library is used so that verifications like fingerprint and face ID are asked before the user can cast their vote. This ensures that only the verified user can cast their vote. The client interface provides a seamless user experience for voters.

[4] Admin Interface - React & Ethers

The admin interface is built using the React framework and ethers.js library. This interface is only accessible by accounts that have deployed the contract. The onlyAdmin modifier in the smart contract allows only authorized addresses to access the UI and add candidates, start the voting portal, and end the voting portal. All the contract communication is done via ethers.js, which is a library used to interact with Ethereum-based blockchains. The admin interface is used to monitor the voting process and ensure the integrity of the election.

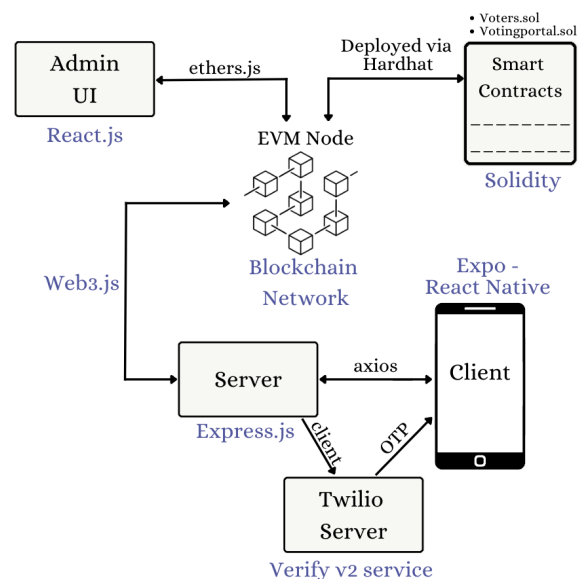


Fig 3: System Flow Block Diagram

V. PROPOSED METHODOLOGY

Prior to the implementation of the proposed system prototype, it is imperative to meticulously document the technology stack, which encompasses an array of frameworks, libraries, and third-party services leveraged. Diverse technologies are employed throughout the entire software development life cycle, ranging from the backend infrastructure to the frontend user interface.

A) DEPLOYING CONTRACT

The first step in the process of our prototype development is contract creation and deployment. We shall write our contracts in solidity and deploy it to Sepolia - an ethereum testnet. We created a metamask account and accumulated some sepolia ethers. We have used Quicknode - a cloud-based infrastructure provider that offers fast and reliable access to blockchain networks. We spin up an ethereum node in the sepolia network. Now using hardhat we configure our deployment process. Hardhat config is provided with the API of the newly created node along with our metamask private keys. Now we shall deploy:

```
sandeep@Sandeeps-MacBook-Air:~$ npx hardhat run scripts/deploy.js --network sepolia
Deploying contracts with account: 0x96F24D8eB318e96bda639e898ab064d6976A287
Account balance: 2038354915603410532
VotePortal address: 0x313d1F6901849FC340a666455aECf7072354A817

sandeep@Sandeeps-MacBook-Air:~$ npx hardhat run scripts/deploy.js --network sepolia
Compiled 1 Solidity file successfully
Deploying contracts with account: 0x96F24D8eB318e96bda639e898ab064d6976A287
Account balance: 2038354915603410532
Voters contract address: 0xF5E12B9C4D3127e208Fab49Ebc3Be78EF40f4565
```

Fig 4: Deploying Contracts

Transactions	Token Transfers (ERC-20)	Contract	Events
17 Latest 1 from a total of 1 transactions			
Transaction Hash	Method	Block	Age
0x8c7d55a86321f4c...	0x4889340	3443864	4 mins ago
Contract Creation			
0 ETH 0.002471331			

Transactions	Token Transfers (ERC-20)	Contract	Events
17 Latest 1 from a total of 1 transactions			
Transaction Hash	Method	Block	Age
0x41391364213863c...	0x609040	3443867	3 mins ago
Contract Creation			
0 ETH 0.00202228			

Fig 5: Deploying Contracts

B) ADMINISTRATOR SETUP

The admin interface is only created to perform 3 functionalities - Adding candidates, starting and stopping voting process. The start and stop process can also be automated, this ensures that the admin does not have any control over the system. The administrator UI is created using React, and ethers.js is used for communicating with the deployed contract and creating transactions via metamask.

E-Voting

This is a blockchain based voting campaign powered by ethereum Admin can connect their wallet and add eligible candidates!

Admin Address: 0x96F24D8eB318e96bda639e898ab064d6976A287

Candidate Name

DATE

Add Candidate

Display All Candidates

Candidate Name: NONE
Vote Count: 0

Candidate Name: ALICE
Vote Count: 0

Candidate Name: BOB
Vote Count: 0

Candidate Name: CANDICE
Vote Count: 0

Start Voting

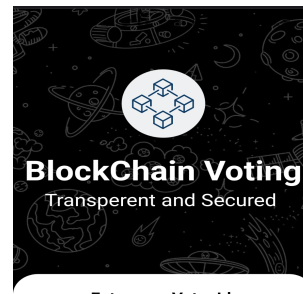
Fig 6: Admin - user interface

C) SERVER CREATION

Once the contract is deployed to the blockchain we establish our server to create endpoints which can communicate with the contract functionality as well as other 3rd party services. We have used node.js as our runtime environment for executing our javascript codes. We have incorporated the express.js framework to spin up our server and installed the web3.js library for smart contract communications. The verify service api from Twilio is used as well for OTP verification.

D) CLIENT LOGIN & VERIFICATION

Once the api endpoints are created and the server is established. We create the client side application using the Expo development tool which is built on top of the React Native framework. Expo provides us with a variety of libraries and functions which make the development process simpler. Redux is used for user state management and axios is used to call the api endpoints. On the client side the login is done via entering the verified voterID registered in the database. The phone number associated with that voterID receives an OTP verification. Once the OTP verification is approved, the user is allowed to vote.



Enter your Voter Id:

1234

Verify Voterid

Enter OTP

885924

Verify OTP

Didn't receive the code? [Resend Code](#)

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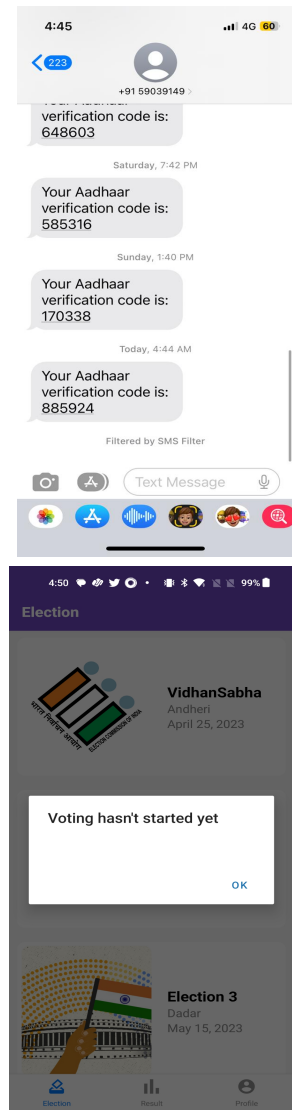
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Fig 7: Client Application - Login & Verification



E) VOTE CASTING

After verification and logging in the client is redirected to a tab navigated screen wherein he can initially see a list of elections inside which are the candidates standing for each election. If the voting process is started users can view the list of candidates and cast their votes if they haven't voted already. On pressing the button next to the candidate the fingerprint sensor or whichever available local authentication process is activated and the user can cast a vote if and only if the authentication matches. Once authenticated the transaction is processed.

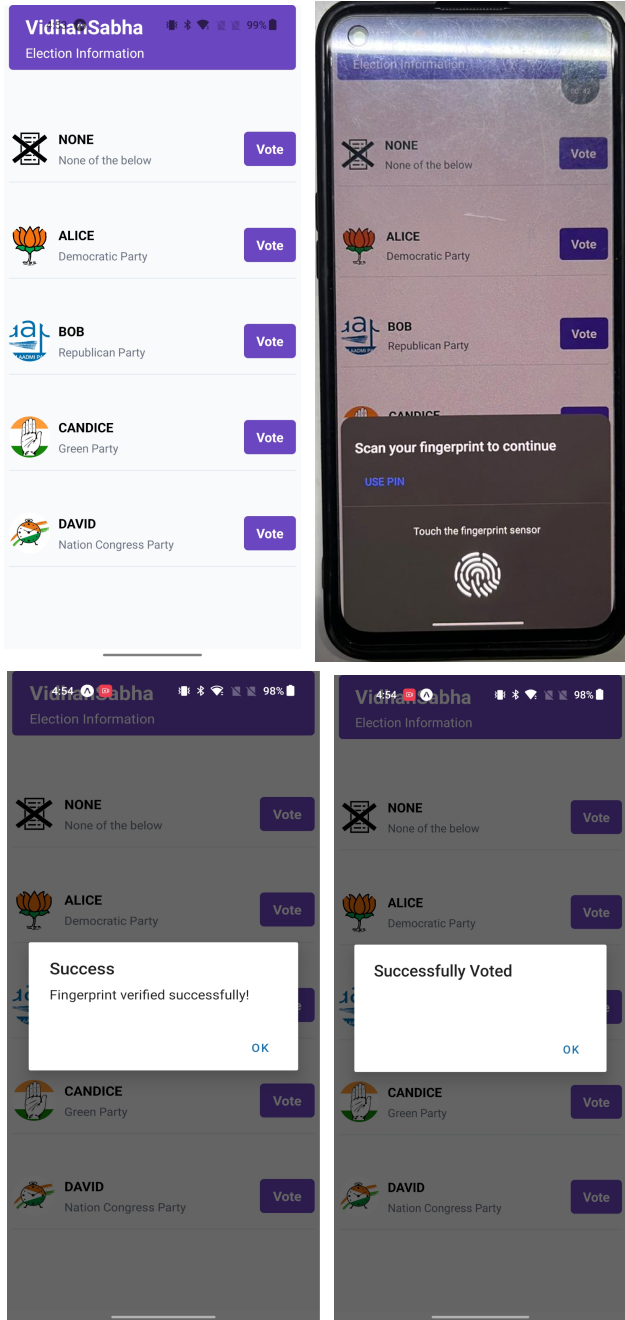


Fig 8: Client Application - Casting Votes

F) RESULT & PROFILE SCREEN

After the voting process is completed, the profile screen indicates the voter's voting status. Users can view the voting results in the form of a bar graph that displays the number of votes obtained by each candidate. The winner of the election is also indicated on the graph.

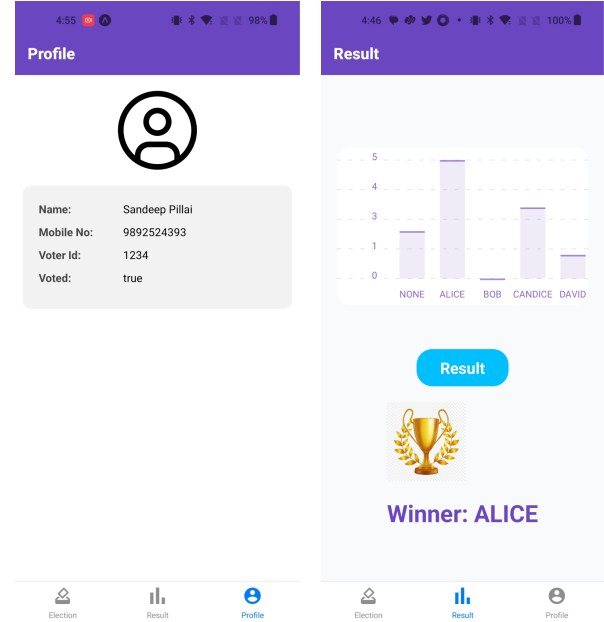


Fig 9: Client Application - Result Screen

VI. RESULTS & ANALYSIS

Analyzing our proposed system and prototype development on blockchain-based e-voting using smartphones, we have successfully implemented a system that offers a fully decentralized and secure client-side app for elections. We demonstrated how our system incorporates dual-layered security with OTP verification at the time of login and fingerprint validation for casting the vote, ensuring the legitimacy of voters. Anonymity is maintained using the SHA3-256 algorithm, which is used to hash the user's voter IDs. Our approach to carrying out transactions using pre-allocated private keys from an array of keys was successful. Hence, our goal of removing dependency on the general public for having knowledge on blockchain and crypto-wallets has been accomplished. The decentralized nature of the app eliminates the risks associated with centralized e-voting systems, while blockchain technology ensures transparency and tamper-resistance. Overall, our app provides a trustworthy and tamper-proof solution for fair and secure elections.

A) ANALYSIS OF ATTACKS

[1] Double Voting Attacks

By mapping voter IDs of users with their voting status in a smart contract and hashing the voter ID, our system effectively mitigates the risk of double voting attacks. Now if an attacker maliciously change's the voting status associated with a voterID, then any such attempt would be rejected by

the network as invalid, as it would not match the agreed-upon rules and protocols of the blockchain.

[2] Voter Anonymity

The use of the SHA3 hashing algorithm to hash user information enhances the anonymity of voters in the context of the blockchain-based e-voting system. However, it is important to note that blockchain transactions are publicly visible, and sophisticated analysis techniques could potentially be used to deanonymize voters based on patterns or other metadata associated with their transactions. Nonetheless, achieving this would require a high level of technical knowledge and expertise, making it a challenging task to accomplish. Therefore, while blockchain technology offers a high level of security and anonymity for e-voting, it is still important to recognize and address potential risks and threats to the system.

[3] Server side attacks

Our system incorporates various security measures to safeguard server resources. One such measure is CORS (Cross-Origin Resource Sharing), which restricts the web application from making requests to a different origin or domain, thereby mitigating the risk of cross-site scripting (XSS) and cross-site request forgery (CSRF) attacks. However, it is imperative to continually monitor and address potential vulnerabilities in the server infrastructure to prevent unauthorized access, data manipulation, or disruption of the voting process. Conducting regular security audits and implementing necessary updates are crucial to maintaining the integrity and reliability of the system.

[4] Client side attacks

Our client-side app features dual security measures that provide robust protection against client-side attacks. The first security measure involves the implementation of OTP verification, which ensures that only authorized individuals sanctioned by the government can access the voting system. This helps to prevent unauthorized users from tampering with or manipulating the voting process, thereby safeguarding the integrity of the electoral system. In addition to OTP authentication, our app employs fingerprint verification, which adds an additional layer of biometric security to authenticate the identity of voters. By combining these two security measures, our system provides a high level of security, ensuring that only legitimate voters can cast their ballots. This approach enhances the integrity and transparency of the voting process, promoting a fair and democratic electoral system.

[5] Admin side attacks

Access to the administrative side of the voting portal is restricted only to authorized accounts who deploy the smart contracts, reducing the risk of admin-side attacks. The admin is however capable of only 3 actions which are to add verified candidates to the system, start the voting process and end the voting process. The start and stop of the voting process however can be automated. Hence the admin truly won't have much authority, still protecting the private key which deployed the contract is still important.

B) PERFORMANCE EVALUATION

[1] Computational Cost

In our app, voters are not required to connect their crypto wallets to pay for transactions. Instead, the server side manages the distribution of load and computation of transactions by selecting a private key from an array of keys. The computational cost associated with these transactions can be assessed by analyzing the gas fees incurred in executing functions that modify the state of the blockchain network. Gas fees represent the cost of computational resources required to perform operations on the blockchain. To evaluate the computational cost of our system, we monitor the gas consumption of various operations, such as candidate registration, vote casting, and updating voter status. By optimizing the smart contract code and minimizing unnecessary computations and variables, we ensure the efficient use of computational resources, thereby reducing the overall cost of transactions. This approach promotes the effective utilization of resources, which is crucial in achieving the scalability and cost-efficiency required for large-scale blockchain-based e-voting systems.

[2] Scalability

The proposed blockchain-based voting app demonstrates scalability through several key features and design considerations. The app utilizes arrays of public keys for transactions, allowing for load distribution across multiple accounts. This approach ensures that the system can handle a higher volume of transactions without significant performance degradation. Additionally, the app is built using React Native, a cross-platform development framework. This enables the app to run seamlessly on both Android and iOS devices, providing wider accessibility and accommodating a larger user base.

In terms of transaction processing, the app eliminates the need for users to connect their personal wallets. Instead, the government takes responsibility for handling the transaction fees associated with voting. This removes any financial burden on the voters and encourages broader participation in the electoral process. To further enhance scalability, future improvements can be made by leveraging advanced scaling solutions such as layer 2 protocols. These solutions can offload transaction processing to secondary layers, alleviating the strain on the main blockchain network and increasing overall scalability.

By incorporating these technical measures and considering scalability from the outset, the blockchain-based voting app is designed to accommodate a growing number of voters and transactions, ensuring a smooth and efficient voting experience for all participants.

VII. FUTURE SCOPE

[1] Improving Transaction Speed with Solana

One of the areas of improvement is the transaction speed and scalability of the blockchain network. By integrating Solana, a high-performance blockchain platform, we can leverage its advanced consensus mechanism and parallel processing capabilities to significantly enhance the speed and throughput of our voting system. This would ensure faster

validation and confirmation of transactions, resulting in a more efficient and responsive voting process.

	Internet Computer	Ethereum	Polkadot	Cardano	Solana	Binance Smart Chain	Zilliqa	Algorand	Avalanche
Average Block Time	0.045 s (1 block)	14 s (1 block)	6 s (1 block)	20 s (1 block)	0.4 s (1 block)	5 s (1 block)	40 s (1 block)	4.5 s (1 block)	2 s (1 block)
Blocks per second	22.5	0.07	0.17	0.05	2.5	0.2	0.02	0.22	0.5
Finality	Web Speed (2 s)	5 min	60 s	2 min	13 s	75 s	2 min	5 s	3 s
TPS	No limit	15	1,000	250	50,000	130	3,000	1,000	4,500 per subnet
Number of Validators	233	6,833	297	2,376	1,027	21	12	100	1027

Fig 10: Different blockchain comparison

[2] Scalability using cloud platforms (AWS)

As the number of users and transactions grows, it becomes crucial to ensure the scalability and reliability of the system. By leveraging Amazon Web Services (AWS), we can take advantage of its cloud infrastructure and services to handle a large number of requests and effectively scale our application. Features such as auto-scaling and load balancing can be implemented to dynamically allocate resources and maintain a smooth and responsive voting system even during peak periods of high user activity.

[3] Implementing Advanced Security Measure

To further enhance the security of the app, advanced security measures can be implemented. This may include the integration of multi-factor authentication methods, such as biometric authentication using facial recognition or iris scanning, in addition to the existing OTP and fingerprint verification. Additionally, robust encryption techniques and secure storage mechanisms can be employed to protect sensitive user data and prevent unauthorized access.

VIII. CONCLUSION

In conclusion, the blockchain-based e-voting system on smartphones presents a promising solution to the challenges and limitations of traditional voting systems. The use of pre-allocated crypto wallet private keys for blockchain transactions, combined with OTP verification and fingerprint authentication, ensures a secure and transparent voting process that is resistant to fraud and manipulation. The system leverages the ubiquity of smartphones and the Ethereum network to provide a decentralized and transparent voting platform that is accessible to people worldwide. The use of Solidity smart contracts ensures the integrity and transparency of the voting process, while the combination of blockchain technology and mobile devices provides a user-friendly and accessible interface for voters. Further research and development are needed to refine and optimize the system and ensure its scalability and reliability in large-scale elections.

IX. ACKNOWLEDGEMENT

We have great pleasure in presenting the research paper on “Blockchain based E-Voting System on Smartphones”. We take this opportunity to express our sincere thanks towards our mentor Prof. D. A. Ambawade, Professor & Head of Department of Electronics & Telecommunication Engineering, S.P.I.T., Mumbai, for providing the technical guidelines and the suggestions regarding this line of work. We would like to express our gratitude towards their constant encouragement, support and guidance throughout the development of the project.

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